

Classification of Epileptic Seizure Using Machine Learning Algorithms

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ABSTRACT

Epileptic seizure detection remains a critical task in medical diagnosis, with machine learning (ML) algorithms offering promising avenues for accurate classification. This study investigates the efficacy of various ML algorithms in classifying epileptic seizures, focusing on the impact of dataset balance and dimensionality reduction techniques. A balanced dataset of seizure and non-seizure cases was utilized, ensuring robust model training across seizure types and frequencies. Feature extraction was performed using multiple techniques, with a particular emphasis on kernel principal component analysis (KPCA) due to its non-linear transformation capabilities. Classification was subsequently achieved through algorithms including k-nearest neighbors (KNN), random forests (RF), support vector machines (SVM), and decision trees (DT). The result obtained from binary classification scenario with SMOTE, showed the highest accuracies with KNN and RF, each achieving 95.14% with KPCA at $d=7$. KPCA yielded the most effective results in producing discriminative features for both binary and multi-class classification, highlighting its value for distinguishing seizure from non-seizure cases. These results indicate that a balanced dataset and an appropriate choice of dimensionality reduction—particularly non-linear KPCA—significantly improve classification performance. These findings support the efficacy of combined feature extraction and machine learning approaches in classifying epilepsy-related cases accurately, underscoring their potential in advancing diagnostic tools for epilepsy management.

Keywords: Epileptic Seizures, Machine Learning, Feature Extraction, Dimensionality Reduction

I. INTRODUCTION

Epilepsy is a prevalent neurological disorder that, although common, can be life-threatening if not properly managed. It impacts individuals of all ages, with seizures triggered by complex chemical changes within the brain's nerve cells. These changes occur within nerve cells composed of positive and negative ions, which are responsible for generating electrical signals Amin et al. [1].

These sudden disruptions can range from mild muscle jerks to severe, generalized, and prolonged convulsions. In addition to impairing movement, this neurological condition affects bowel and bladder control, alters

consciousness, and interferes with cognitive functions Sharmila et al. [2].

Epilepsy and epileptic seizures affect a diverse population, with approximately 70% of cases occurring in adults and 30% in children Farooq et al [3]. In 70% of cases, the underlying cause of epilepsy in both adults and children remains unknown. Clinically, seizures are classified as epilepsy when they occur recurrently. There are two primary classifications of seizures: partial (focal) and generalized Sharmila et al. [2]. In a partial seizure, only a specific region of the brain is affected, while generalized seizures involve the entire brain, state Fisher [4].

Epilepsy progresses through four key stages: interictal, preictal, ictal, and postictal. The most well-established method for detecting abnormal seizures is through the use

of an electroencephalogram (EEG), which involves placing electrodes on the scalp either invasively or non-invasively to measure electrical activity in the brain Amin et al. [1]. Detection techniques for epileptic seizures using EEG data vary depending on the specific characteristics of the dataset Sharmila et al [2]. Accurate feature extraction is critical for improving the prediction and classification of epileptic seizures, thereby enhancing the precision of the results.

Significant research efforts have focused on EEG analysis for feature extraction, which plays a vital role in detecting and predicting various brain disorders Harpala & Bairagi, 2021 [5]. Much of this research aims to differentiate normal seizures from epileptic ones. Unlike epileptic seizures, which are triggered by abnormal brain activity, non-epileptic seizures can be induced by emotional stress or trauma Lafrance et al. [6]. Advanced methods such as wavelet transforms and hypothetical testing for feature refinement are being applied to enhance classification accuracy Thara et al. [7]. The computational complexity of classifiers can be reduced by selecting the most relevant features. Recent studies are increasingly focusing on the prediction of epileptic seizures, which remains a challenging area of research Boonyakitanont [8].

II. RELATED WORKS

This section is dedicated to examining the contributions of various researchers who have explored the prediction of epileptic seizures using machine learning techniques. A review of recent studies focusing on the diagnosis of epileptic seizures through the analysis of EEG signals is presented, highlighting key advancements and methodologies in the field.

In the study conducted by Almustafa [9], various machine learning techniques were employed for the classification of an epileptic seizure dataset using multiple classifiers. The results demonstrated that the Random Forest classifier outperformed other methods, including K-Nearest Neighbors (K-NN), Naïve Bayes, Logistic Regression, Decision Tree (D.T.), J48, and Stochastic Gradient Descent (SGD), achieving an accuracy of 97.08%, an ROC value of 0.996, and an RMSE of 0.1527. Sensitivity analysis was performed on several of these classifiers to evaluate their performance in classifying the

epileptic seizure dataset under varying parameter settings. Additionally, feature selection based on attribute variance was utilized to improve the predictive capability of the dataset.

A study conducted by Pippa et al [10] aimed to classify epileptic and non-epileptic EEG events by evaluating three machine learning methods on data from 11 subjects. In an inter-subject cross-validation setting, the Bayesian Network method achieved the highest accuracy, reaching up to 86%. The other methods tested were Random Committee, which achieved 83% accuracy, and Random Forest, with 74% accuracy. The study concluded that the Bayesian Network method demonstrated sufficient generalizability, indicating its potential to assist in the diagnosis of epilepsy.

Nandy et al [11] employed a Support Vector Machine (SVM) classifier for the classification of epileptic seizures, utilizing Bayesian optimization to enhance the SVM hyper-parameters. For comparative analysis, they also applied Linear Discriminant Analysis (LDA) and Quadratic Linear Discriminant Analysis (QLDA). The results indicated that the SVM classifier achieved an accuracy of 97.05%.

In the study conducted by Usman [12], the authors proposed a system that employs Empirical Mode Decomposition (EMD) to extract features from EEG signals, utilizing both time and frequency domain characteristics. The CHB-MIT dataset was used for evaluation. The model demonstrated a high true positive rate of 92.23%, with an average prediction time of 23.60 minutes across all subjects in the dataset.

Hamad et al. [13] utilized the Discrete Wavelet Transform (DWT) approach to extract features from EEG signals, which were subsequently employed to train a Support Vector Machine (SVM) using a radial basis function (RBF) kernel. To enhance the effectiveness of EEG classification, the Grey Wolf Optimizer (GWO) was implemented to select a relevant subset of features and optimize the SVM parameters. The experimental results demonstrated that the proposed GWO-SVM approach achieved a remarkable classification accuracy of 100% in detecting epileptic seizures, thereby improving the diagnostic capabilities for epilepsy. Furthermore, the GWO-SVM approach was compared with Genetic Algorithm (GA) integrated with Support Vector Machines (GA-SVM) and standard SVMs using the RBF kernel. The computational results indicated that the

GWO-SVM approach outperformed both GA-SVM and conventional SVMs in terms of classification accuracy.

In the study conducted by Nahzat et al [14], the authors aimed to employ Principal Component Analysis (PCA) as a feature reduction technique to identify the optimal classification algorithm for epileptic seizures. Various algorithms, including K-Nearest Neighbors (KNN), Random Forest (RF), Support Vector Machine (SVM), Artificial Neural Network (ANN), and Decision Tree (DT), were utilized to predict epilepsy, with the performance of the classifiers evaluated both with and without PCA. The results indicated that when PCA was applied, the Random Forest classifier achieved the highest accuracy of 97% with low computational times for both training and testing. Furthermore, K-Nearest Neighbors and Random Forest achieved an accuracy of 99% without the application of PCA, demonstrating superior performance compared to other machine learning techniques.

In 2019, Usman et al. [15] employed Principal Component Analysis (PCA) for feature extraction in conjunction with a Support Vector Machine (SVM) classifier for the classification of epileptic seizures. The proposed model demonstrated an average sensitivity of 93.1% in its performance Usman et al[15].

In 2016, Sharmila and Geethanjali [2] utilized Discrete Wavelet Transforms (DWT) to decompose EEG data into distinct sub-bands, subsequently deriving statistical features for analysis. The classifiers were trained using these DWT-derived statistical features, and the EEG signals were then classified as either epileptic or non-epileptic using two classifiers: K-Nearest Neighbors (KNN) and Naive Bayes. The study compared the effectiveness of 14 different two-class combinations for epilepsy detection. The experimental results indicated that the Naive Bayes classifier achieved the highest accuracy with reduced computational time for most dataset combinations in diagnosing epileptic seizures Sharmila & Geethanjali [2].

A study aimed at classifying epileptic and non-epileptic EEG events evaluated three machine learning methods across 11 subjects. The findings revealed that, within an inter-subject cross-validation framework, one of the evaluated methods achieved an accuracy of up to 86%. The Bayesian Network method demonstrated the highest accuracy among the classifiers tested, outperforming the Random Committee, which achieved an accuracy of 83%,

and the Random Forest, which attained 74% accuracy. The study concluded that the Bayesian Network method exhibits satisfactory generalization capabilities, thereby facilitating the diagnosis of epilepsy (E. Pippa et al., 2014) [10].

Shafiezadeh, et al [16] focused on a cross-patient perspective, addressing the complexities arising from EEG variability across different subjects. A comparative analysis was conducted on pattern recognition algorithms utilized for EEG-based identification of epileptic seizures, specifically examining the performance of Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) classifiers. The results indicated that both methods exhibited comparable performance; however, KNN demonstrated slightly higher accuracy under certain conditions.

Another patient non-specific classifier demonstrated commendable performance in this study, which focused on seizure types that exhibit minimal variability among different patients. The classifier employed for this research was the Support Vector Machine, utilizing EEG recordings from 57 patients, totalling approximately 43 hours of data. The findings indicated that incorporating seizure morphology significantly enhances detection performance. The detection achieved reliability with a low false alarm rate, resulting in an average correct detection rate exceeding 96% (Ralph Meier et al., 2008) A study aimed at identifying a classifier for personalized seizure detection employed K-Nearest Neighbor (KNN) and Support Vector Machines (SVM) as the classification methods. Both approaches achieved accuracies exceeding 80%, with KNN demonstrating slightly superior performance compared to SVM. Notably, KNN exhibited an onset sensitivity of 100% when evaluated on a cohort of ten patients Adam Page et al [17].

In their study, Usman et al. proposed a system that employs empirical mode decomposition (EMD) to extract time and frequency domain features from EEG signals. Utilizing the CHB-MIT dataset, the model achieved a high true-positive rate of 92.23%, with an average prediction time of 23.60 minutes for all subjects in the dataset.

In their research, Serna et al [18] proposed a system centered on feature extraction and classification. The model utilized Taylor Fourier transforms, rhythm-specific methods, and filter banks for preprocessing, followed by feature extraction and classification through

a Support Vector Machine (SVM). This approach yielded an accuracy of 94.88%, utilizing the Bonn University Database for evaluation.

Swami et al [19] employed the dual-tree complex wavelet transform (DTCWT) for the decomposition of signals, subsequently calculating various statistical measures. These statistical features were then utilized to train a General Regression Neural Network (GRNN) classifier. The proposed model demonstrated an accuracy of 95.24%.

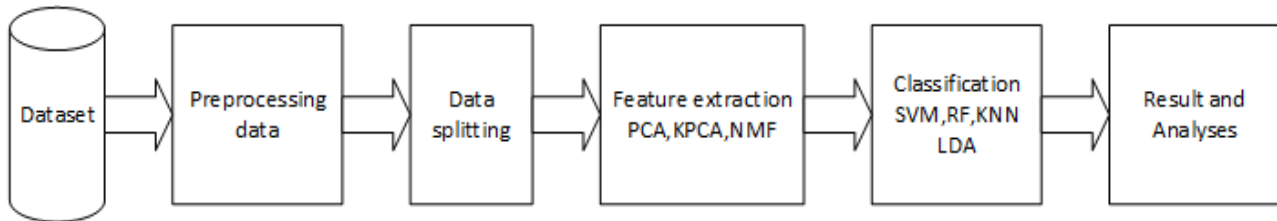


Figure 3.2. proposed model

Analysis of the Proposed System

The proposed system architecture includes:

1. Input Layer: Raw EEG signals acquired from patients.
2. Preprocessing Module: Noise reduction and normalization techniques applied to enhance signal quality.
3. Feature Extraction Module: Utilization of advanced methods such as Principal Component Analysis (PCA), Kernel Principal Component Analysis (KPCA) and Non-Negative Matrix Factorization (NMF) to extract discriminative features from pre-processed signals.
4. Classification Module: Implementation of various ML algorithms (e.g., SVM, Random Forest) for effective classification based on extracted features.
5. Output Layer: Final classification results indicating seizure presence or absence.

1) System Design

The primary objective of this study is to identify the most effective classification algorithm for epileptic seizure detection by employing the Principal Component Analysis (PCA) feature reduction technique on the dataset. The subsequent sections will discuss the various phases of the proposed methodology in detail. Figure 3.2

III. METHODS AND MATERIAL

The architecture of the proposed system consists of several interconnected components designed to streamline the process from raw EEG signal acquisition to seizure classification.

provides a visual representation of the overall flowchart of the suggested model.

Dataset Description

The EEG dataset used in this study is sourced from a publicly available repository, Kaggle.com. It comprises five distinct classes, labeled 1 through 5, with each class containing 100 signals of 23.6 seconds in duration. Signals recorded from five healthy individuals are represented by classes 4 and 5.

with Class 5 captured while their eyes were open and Class 4 while their eyes were closed. The remaining three classes (1, 2, and 3) correspond to recordings from five epileptic patients. Specifically, Class 3 was recorded from the hippocampal contralateral hemisphere during the pre-seizure state, Class 2 from the epileptogenic region of the patient without seizure activity, and Class 1 during actual seizure events.

A 128-channel system with a sampling rate of 173.61 Hz and a 12-bit analog-to-digital converter were used to record all EEG signals. The dataset consists of 11,500 samples, each comprising 178 attributes, and is evenly distributed across the classes. Importantly only people in class 1 have ever had an epileptic seizure, whereas people in classes 2, 3, 4, and 5 have never had one (Andrzejak et al., 2001).

Consequently, for the purpose of our analysis, we transform the dataset into a binary classification problem: epileptic versus non-epileptic seizures. To achieve this, classes 2, 3, 4, and 5 are merged into a single class representing non-epileptic seizure patients.

Table 1 provides the number of instances in each class prior to merging, indicating an equal distribution of samples. Figure 2 presents a visual sample from the epileptic seizure dataset.

Table 1 Description of the Dataset and the Total Number of Cases in Every Class

Class	Class Description	Patient State	Number of Cases	Binary Classification
1	Seizure activity recorded from epileptic patients	General epilepsy (with seizures)	2300	2300 (Seizure)
2	EEG recorded from the epileptogenic region (no seizure activity)	Partial epilepsy (without seizures)	2300	9200 (No Seizure)
3	EEG recorded from a healthy brain region of epileptic patients	Partial epilepsy (without seizures)	2300	
4	EEG recorded from healthy individuals with eyes closed	Healthy	2300	
5	EEG recorded from healthy individuals with eyes open	Healthy	2300	

2) Data Preprocessing

Data preprocessing is a crucial and necessary phase in machine learning workflows, as it directly impacts the accuracy, reliability, and overall performance of predictive models Soni, [20]. This process involves transforming raw, unstructured data into a format suitable for analysis and classification, ensuring the effectiveness of machine learning algorithms applied to the dataset.

Real-world data often suffers from incompleteness, inconsistency, and the absence of discernible patterns, in addition to containing various errors. To address these challenges, data preprocessing serves as a proven approach to refine and prepare raw data, ensuring it is structured and suitable for subsequent analysis and machine learning processes.

In the dataset utilized for this study, there are no missing values (NaN). However, as indicated in Table 1, an imbalanced class distribution exists in the binary classification. To address this issue and ensure more reliable model performance, oversampling techniques were employed to balance the dataset.

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Synthetic Minority Oversampling Technique (SMOTE)

A common technique for addressing class imbalance in datasets is SMOTE, or the Synthetic Minority Oversampling Technique. This is especially useful in classification problems where the minority class is underrepresented in comparison to the majority class.

. Introduced by Chawla et al [21], By interpolating between minority instances that already exist, SMOTE creates synthetic instances of the minority class. Sets This is achieved by selecting a random sample from the minority class and creating new samples along the line segments joining this sample and its nearest neighbors.

Unlike traditional oversampling methods that involve duplicating existing minority class instances, SMOTE reduces the risk of overfitting by introducing variability in the generated data. The technique constructs synthetic data points, which enhances the diversity of the training set and leads to better generalization of machine learning models. As a result, classifiers trained on SMOTE-enhanced datasets tend to have improved performance, especially in scenarios where the imbalance would otherwise cause the model to bias toward the majority class.

SMOTE is particularly effective when combined with algorithms sensitive to class distributions, such as decision trees and support vector machines, and is often applied in domains like fraud detection, medical diagnostics, and anomaly detection. However, it is important to carefully assess the appropriateness of

SMOTE for a given problem, as generating synthetic instances can sometimes lead to the introduction of noise if not applied judiciously [22].

Adaptive Synthetic Sampling (ADASYN)

In order to address class imbalance, He et al [23] presented the Adaptive Synthetic (ADASYN) technique, which adaptively generates minority data instances according to their distribution. The main concept of ADASYN was to automatically calculate the number of synthetic instances that must be created for each minority data instance using density distribution as a criterion. ADASYN provided different weights for different minority instances. The ADASYN approach was more efficient than other algorithms that provided different weights for different minority instances because it adaptively updated the distribution based on the characteristics of its data distribution.

Feature Extraction

The technique of extracting unique and significant information from unprocessed data is known as feature extraction. It is essential to reducing the dimensionality of the dataset while retaining the attributes that are most essential to the classification procedure.

The objective for extracting the features at this stage is to reduce the dimensionality of the dataset by generating a fresh set of attributes derived from the existing features. This reduced feature set should retain the majority of the important data contained in the initial dataset. By combining and transforming the initial attributes, a more concise representation of the key features is obtained, which effectively summarizes the underlying information Ippolito [24].

As previously mentioned, the dataset contains 178 features. Utilizing all these features for training would result in prolonged training times. Therefore, PCA, or principal component analysis, was used in this study alongside KPCA and NMF were employed for feature extraction and dimensionality reduction to enhance computational efficiency while preserving the most significant information for classification.

3) Algorithms used for Classification

The classification and predictive machine learning algorithms implemented in this study include the following:

- i. Linear Discriminant Analysis
- ii. Decision Tree
- iii. Random Forest
- iv. K-Nearest Neighbor
- v. Support Vector Machine
- vi. Naïve Bayes
- vii. Logistic Regression

4) Evaluation methods

The effectiveness of machine learning models, encompassing both classification and regression techniques, is quantitatively assessed using evaluation metrics, commonly referred to as loss functions. In the context of machine learning, most classification problems entail the categorization or labeling of specific test instances. This process is achieved by employing a function that has learned to map distinct output variables or classes to their corresponding discrete input samples.

Binary classification problems are characterized by having only 2 types of observations within the dataset. In contrast, multi-class classification challenges involve the assignment of data points to more than two classes. The effectiveness of any classification model is gauged by its ability to accurately predict the classes associated with data points. Consequently, evaluation metrics serve as performance assessment tools that measure the accuracy of model predictions. The results of these evaluations inform the choice of the most successful learning model for addressing a specific classification or regression task.

Accuracy: One essential parameter for assessing a machine learning model's overall performance is accuracy. It shows the proportion of occurrences that were properly classified to all instances that were examined.

In epilepsy seizure detection, accuracy can be calculated as follows:

$$Accuracy = \frac{TP + TN}{FP + FN + TP + TN}$$

Where:

- The number of accurately classified seizure events is represented by True Positives (TP).
- The number of accurately categorized non-seizure cases is represented by True Negatives (TN).

- The number of non-seizures that are mistakenly labeled as seizures is known as False Positives (FP).
- The quantity of seizure cases that are mistakenly labeled as non-seizures is known as False Negatives (FN).

Confusion Matrix: provides a summarizing of the number of true positive, true negative, false positive, and false negative predictions to provide a thorough analysis of the model's performance. It is especially helpful for comprehending the kinds of mistakes the model makes. Several evaluation metrics can be derived from it, including:

- True Positive Rate or Recall (also known as Sensitivity): Indicates the percentage of actual positive cases that the model accurately detected.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

- True Negative Rate (Specificity): Indicates the percentage of real negative cases that the model accurately detected.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- Precision (also known as positive predictive value): Indicates the percentage of actual positive predictions among all of the model's positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- F1 Score: A balanced indicator of precision and recall, expressed as the harmonic mean of the two.

$$F1 \text{ Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

IV. RESULTS AND DISCUSSION

In this section, we discussed the outcomes of the epilepsy classification problem combining different data processing techniques and machine learning algorithms. The dataset for this project is made of one hundred and seventy eight (178) features labeled X1 to X178. There are eleven thousand five hundred (11500) data points (observations) in the dataset. Furthermore, there are five

(5) class labels for the different epileptic categories in the dataset. In this study, we represent the data points as X and the class labels as y . Therefore, dataset size can be expressed as $X = \mathbf{R}^{11500 \times 178}$ and $y = \mathbf{R}^{11500 \times 1}$. The different epilepsy classes are labeled 1, 2, 3, 4 and 5. In this case, members of 2, 3, 4 and 5 labels has never experienced an epileptic seizure, whereas members of class 1 had experienced an epileptic seizure. On the other hand, members of class 1 had experienced an epileptic seizure. In this study, we performed the epileptic seizure classification based on two scenarios:

1. as a multi-class classification problem

2. as a binary classification problem

The original representation (format) of the dataset indicated that it is a multi-class classification task. However, in this work, we combined classes 2, 3, 4 and 5 into a single class labeled 0, since members of those classes had never experienced an epileptic seizure. By combining these classes (2, 3, 4, and 5), we have reorganized the data into a binary classification problem with two classes 0 and 1. Although, the sizes of the dataset for both the multi-class and binary classification problems are the same, however there is imbalance data points for the binary classification problem. For the binary classification problem, the number of data points for the majority class 0 is 9200 and that of the minority class 1 is 2300. For this reason, there is need to increase the size of the minority data points using oversampling techniques such as SMOTE and ADASYN. This step will help to create a balanced dataset that improves the accuracy of the classification models while minimizing overfitting.

After obtaining a balanced dataset for both the binary and multi-class classification problems, we applied the StandardScaler class in Scikit-learn preprocessing module to normalized the data since the features values may varies as a result of different measurement method. Subsequently, we performed PCA, NMF and KPCA dimensionality reduction techniques on the balanced and normalized dataset in order to extract discriminative features that will enhance the data categorization task and allows few input values to the classification models. The goal is to reduce the original feature dimension of the data from 178 to a smaller dimension represented as $\mathbf{d}$, for both the multi-class and binary classification tasks. The extracted discriminative features serve as input to the

different machine learning algorithms. The machine learning models applied here are Random Forest (RF), Logistic Regression (LR), K-Nearest-Neighbor (KNN), support vector machine (SVM), Naïve Bayes (NB) and Linear Discriminant Analysis (LDA).

First, we present the baseline accuracy scores for each model mentioned above as a way of performance evaluation and comparison. Thereafter, we tuned the individual model parameters with the aim of achieving a better classification results.

Table 4.3 Binary classification accuracies using discriminative features with n_neighbors=3 for the ADASYN oversampling algorithm and feature dimension d=7

Models	PCA with Tuned model Parameters	PCA with Default model Parameters	NMF with Tuned model Parameters	NMF with Default model Parameters	KPCA with Tuned model Parameters	KPCA with Default model Parameters
RF	84.57	93.77	77.11	92.26	90.04	95.14
KNN	95.01	95.01	91.39	91.39	95.14	95.14
CART	90.39	90.12	87.14	86.86	92.66	92.93
NB	84.40	84.40	68.29	68.29	84.72	84.72
SVM	86.88	92.05	70.64	86.69	88.73	93.68
LR	56.52	56.52	56.74	56.74	87.20	86.49
LDA	56.57	56.57	56.78	56.79	74.02	74.02

Tables 4.3 shows that, the KPCA data representation when used for training and evaluating the various model performance, produced the highest classification accuracy score of 95.14% for both tuned and default KNN model parameters. The RF classifier produced the same highest accuracy score of 95.14% using the default model parameters as shown in Table 4.3. Tuning the RF model parameters did not improve the model performance in this case. The number of nearest neighbor considered in order to determine the new synthetic data points using ADASYN sampling method was three (3). We increased the number of nearest neighbor to five (5) and performed the same classification task and the result indicated that the KNN classifier produced the highest scores for

accuracy of 95.05% in both cases (default and tuned model parameters). This suggests that, a non-linear discriminative data representation of the seizure dataset can help categorize a seizure case from a non-seizure case. Furthermore, with only seven (7) KPCA discriminative features, we can account for 95.14% classification accuracy in distinguishing between a seizure and non-seizure cases using KNN classifier. Subsequently, we increased the dimension of the discriminative features, before training and evaluating the performance of the models. We noticed that, increasing the number of discriminative attributes improved the models classification accuracy. However, the increment was not significant.

Table 4.5 Binary Classification accuracies using discriminative features with K_neighbors=3 for the SMOTE oversampling algorithm and d=7

Models	PCA with Tuned model Parameters	PCA with Default model Parameters	NMF with Tuned model Parameters	NMF with Default model Parameters	KPCA with Tuned model Parameters	KPCA with Default model Parameters
RF	91.76	95.93	76.96	94.83	91.89	95.41
KNN	96.07	96.24	92.57	92.57	95.97	95.98
CART	92.02	93.07	90.93	90.52	92.50	92.65

NB	93.37	93.52	73.13	73.13	93.11	93.11
SVM	89.46	94.24	72.61	88.74	89.52	94.13
LR	50.20	49.61	47.70	47.70	49.46	49.46
LDA	50.22	49.61	47.70	47.70	50.70	50.70

Similarly, as illustrated in Tables 4.5, the classifiers accuracy scores when the imbalanced binary dataset was balanced using the SMOTE oversampling method. Again, the KPCA feature representation produced the highest accuracy scores of 95.98% when the KNN classifier was trained with the features.

As evidenced in Figures 4.1, 4.2 and 4.3, represents the confusion matrices for the KNN, SVM and RF classifiers,

when the balanced binary dataset is generated using SMOTE method ($k_neighbors=5$), and the models are trained and evaluated using the PCA discriminative features. The accuracy scores produced by the KNN, SVM and RF learning algorithms are 96.46%, 94.67 and 94.25% respectively. The scores show that, the KNN classifier produce the best classification accuracy score of 96.46%.

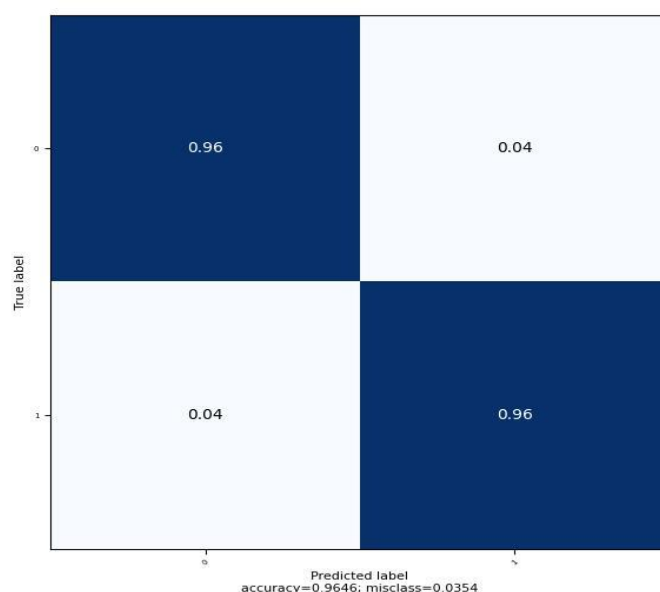


Figure 4.1 KNN confusion matrix with PCA and SMOTE oversampling method

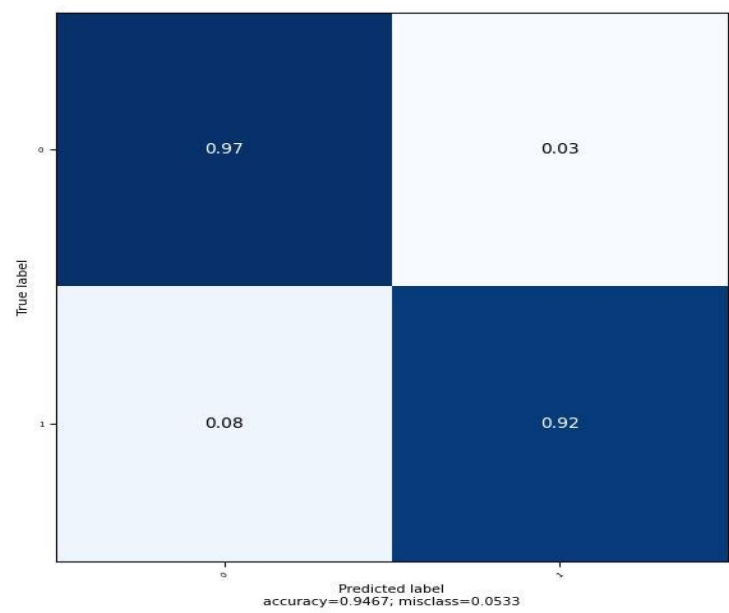


Figure 4.2 SVM confusion matrix with PCA AND SMOTE oversampling method

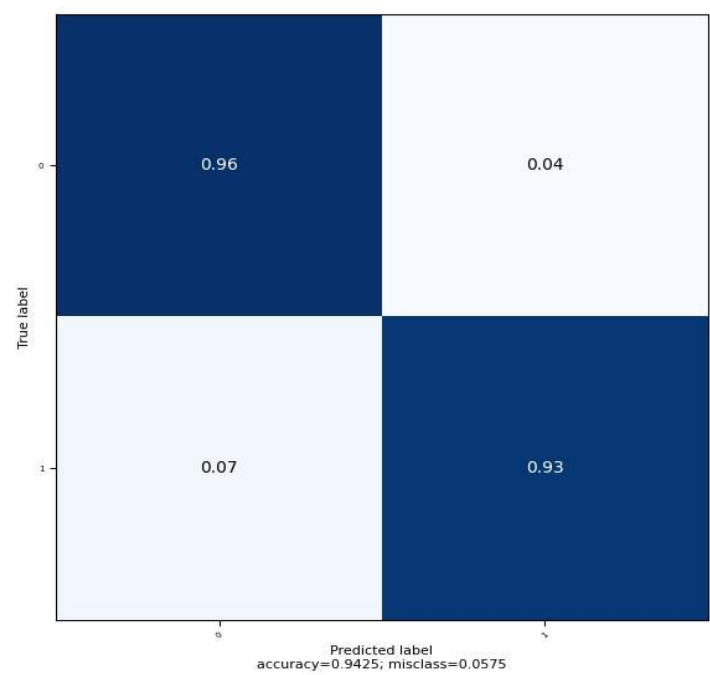


Figure 4.3 RF confusion matrix with PCA and SMOTE oversampling method

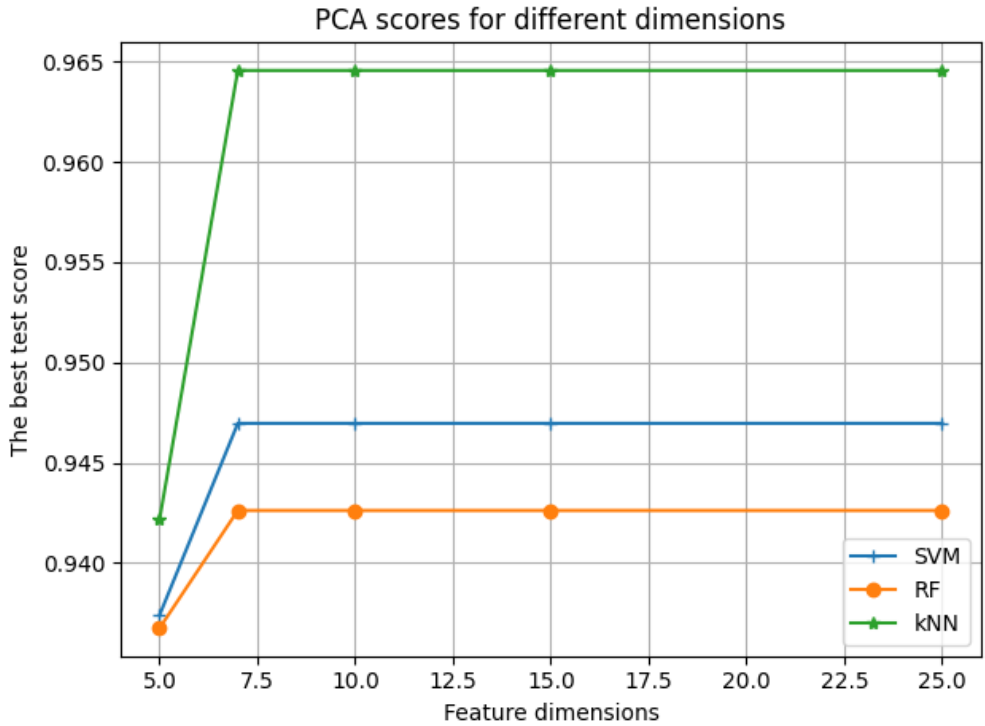


Figure 4.4 Accuracy scores for different classifiers using PCA representation with different dimensions

Again, the KNN classifier produce the best classification accuracy score of 96.22% as also illustrated in Figure 4.8. Figure 4.8 donates the classification accuracies scores of the KNN, SVM and RF algorithms when different dimension of the KPCA discriminative features was used to train and evaluate the classifiers performance. The results suggest that, a non-linear dimensionality reduction technique like KPCA with kernel parameter equals “rbf”

can be used to produce discriminative features that can catigorise the seizure dataset into seizure and non-seizure cases. As seen in Figures 4.12, the KNN, SVM and RF classification accuracies are 93.26%, 93.41% and 89.19% respectively as also shown in Figure 4.12. The results show that, the NMF discriminative feature can be used to categorise a seizure case for a non-seizure case.

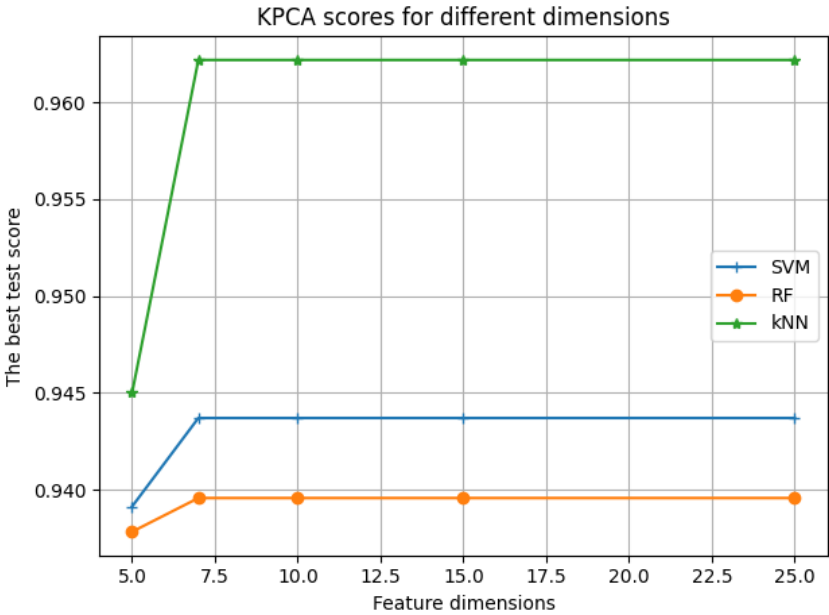


Figure 4.8 Accuracy scores for different classifiers using KPCA representation with different dimensions

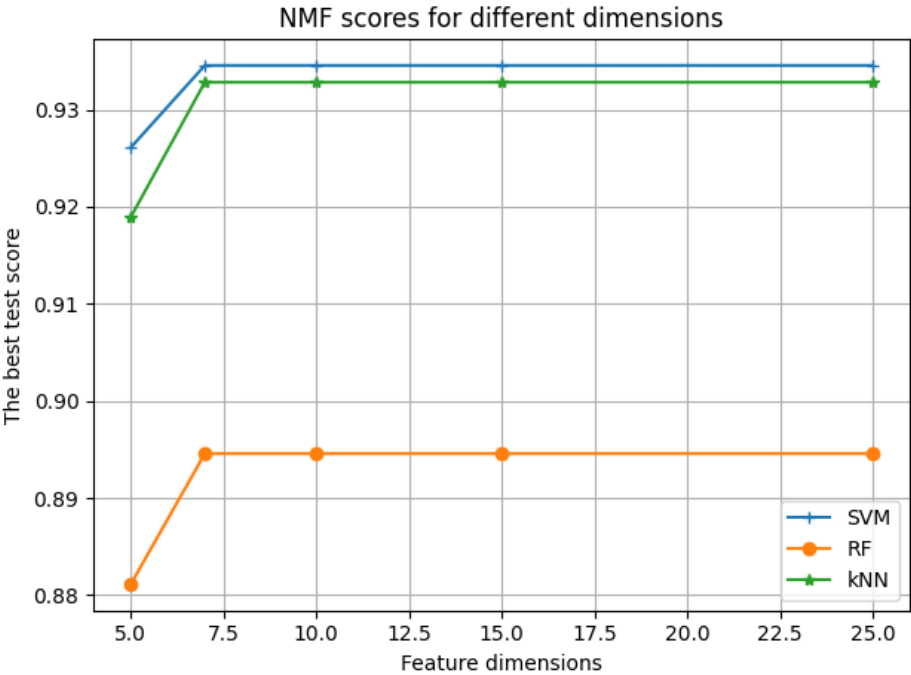


Figure 4.12 Accuracy scores for different classifiers using NMF representation with different

V. CONCLUSION

In conclusion, the results indicate that a balanced dataset and an appropriate choice of dimensionality reduction (particularly non-linear KPCA) significantly improve

classification performance. These findings support the efficacy of combined feature extraction and machine learning approaches in classifying epilepsy-related cases accurately.

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